# Functions and arguments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

## One connected function-and-argument record.

Explain the definition, trace calls, repair fixed constants and state the input assumptions.

**Keep one habit:** Keep contract, parameter mapping, call order, returned/printed distinction and observed evidence together.

MAIN LESSON · SECTION 1

## Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Try the re**call** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

### Lists and indexing

> **My remembered idea, example and correction**

### Branching and boundary tests

> **My remembered idea, example and correction**

### Values variables and types

> **My remembered idea, example and correction**

## W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Main printing function show_remaining(capacity,used) assumes nonnegative integers with used<=capacity. W2 calls(8,3),(8,8),(8,0)print5,0,8. Definition alone no output; missing required argument raises TypeError. Faulty body 12-4 ignores inputs. Separate picture show_free(11,3)/(3,11)prints 8/-8 to show arithmetic order, not valid negative free-space.

  Return preview hands numeric value to caller; outer print displays it.

- In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 6

## W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Main printing function show_remaining(capacity,used) assumes nonnegative integers with used<=capacity. W2 calls(8,3),(8,8),(8,0)print5,0,8. Definition alone no output; missing required argument raises TypeError. Faulty body 12-4 ignores inputs. Separate picture show_free(11,3)/(3,11)prints 8/-8 to show arithmetic order, not valid negative free-space.

  Return preview hands numeric value to caller; outer print displays it.

- Reference for this exercise: open this week's student lesson, "Functions and arguments", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 7

## W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Main printing function show_remaining(capacity,used) assumes nonnegative integers with used<=capacity. W2 calls(8,3),(8,8),(8,0)print5,0,8. Definition alone no output; missing required argument raises TypeError. Faulty body 12-4 ignores inputs. Separate picture show_free(11,3)/(3,11)prints 8/-8 to show arithmetic order, not valid negative free-space.

  Return preview hands numeric value to caller; outer print displays it.

- Reference for this exercise: open this week's student lesson, "Functions and arguments", at "Find and repair a mistake". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 9

## W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Main printing function show_remaining(capacity,used) assumes nonnegative integers with used<=capacity. W2 calls(8,3),(8,8),(8,0)print5,0,8. Definition alone no output; missing required argument raises TypeError. Faulty body 12-4 ignores inputs. Separate picture show_free(11,3)/(3,11)prints 8/-8 to show arithmetic order, not valid negative free-space.

  Return preview hands numeric value to caller; outer print displays it.

- The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 4

## Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Predict both returned and printed values, showing which **argument** supplies first and second. What is displayed if the program contains only the **function definition** and no **call**s?

  Why does swapping these **argument**s change the result?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Independent case: Match positional arguments to parameters

```
def difference(first, second):
    return first - second
print(difference(11, 4))
print(difference(4, 11))
```

> **My independent explanation, trace and limits**
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MAIN LESSON · SECTION 10

# Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

## Task and evidence record

- Add a labelled **function**s and **argument**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish a **function definition** from a **function call**.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

**My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

# Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

## Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Use W2: Follow the execution order for **Function**s and **argument**s. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

# Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

## Task and evidence record

- Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

# Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

- Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

  Predict each **output** and explain why show_stock(2, 3) is a different case.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

## Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

### Task and evidence record

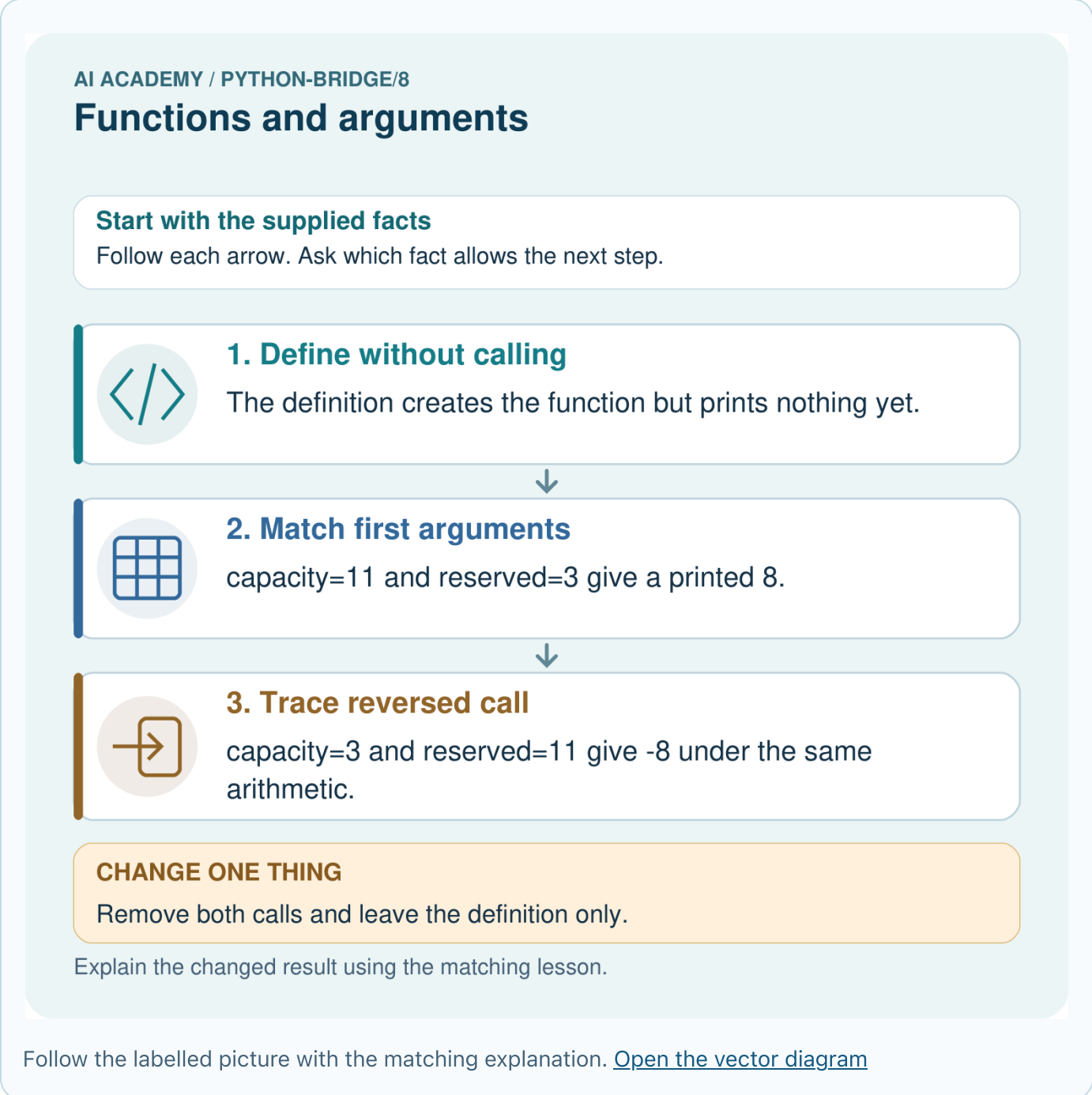
- Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces.

- Python 3: def show_free(capacity,reserved): print(capacity-reserved). **Call**s: show_free(11,3); show_free(3,11).

- Match every **argument** to a **parameter** for the two **call**s. Why do **output**s differ, and what prints when both **call**s are removed?

- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.




Follow the labelled picture with the matching explanation. Open the vector diagram

**My picture-based explanation and changed-case prediction**

▼ **Reference · Project portfolio milestone**

REFERENCE · SECTION 13

# Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

## Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a **function** on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

| Evidence to keep | My record |
| --- | --- |
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |

| Evidence to keep | My record |
| --- | --- |
| One error or limit and the next test | |

**Explain what this week adds to your project. Reuse last week's record rather than starting again.**

**▼ Extra practice · Predict, check and repair**

REFERENCE · SECTION 14

# Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

## First question and hint

- Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces.
- Which labelled step matches this detail? capacity=3 and reserved=11 give -8 under the same arithmetic.
- 1. Match first **argument**s
- 2. Trace reversed **call**

- 3. Define without **call**ing

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

**▼ Check the first explanation**

After your attempt, compare with the supported explanation: Trace reversed **call** capacity=3 and reserved=11 give -8 under the same arithmetic. Positional **argument**s match **parameter** names in order.

A **function definition** is not an execution of its body; separate **call**s supply separate **input** values.

## Changed-case question and hint

- Changed-case practice: Remove both **call**s and leave the **definition** only. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. We can decide the result without examining the changed **input** or the procedure.

- 3. No **output** is printed.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: No **output** is printed. The body runs only when the **function** is **call**ed.

## Repair and keep the record

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions.

Predict all original **output**s, then name the first **output** and stopping point after using index 3.. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Predict, run and debug**

**Keep your completed work**

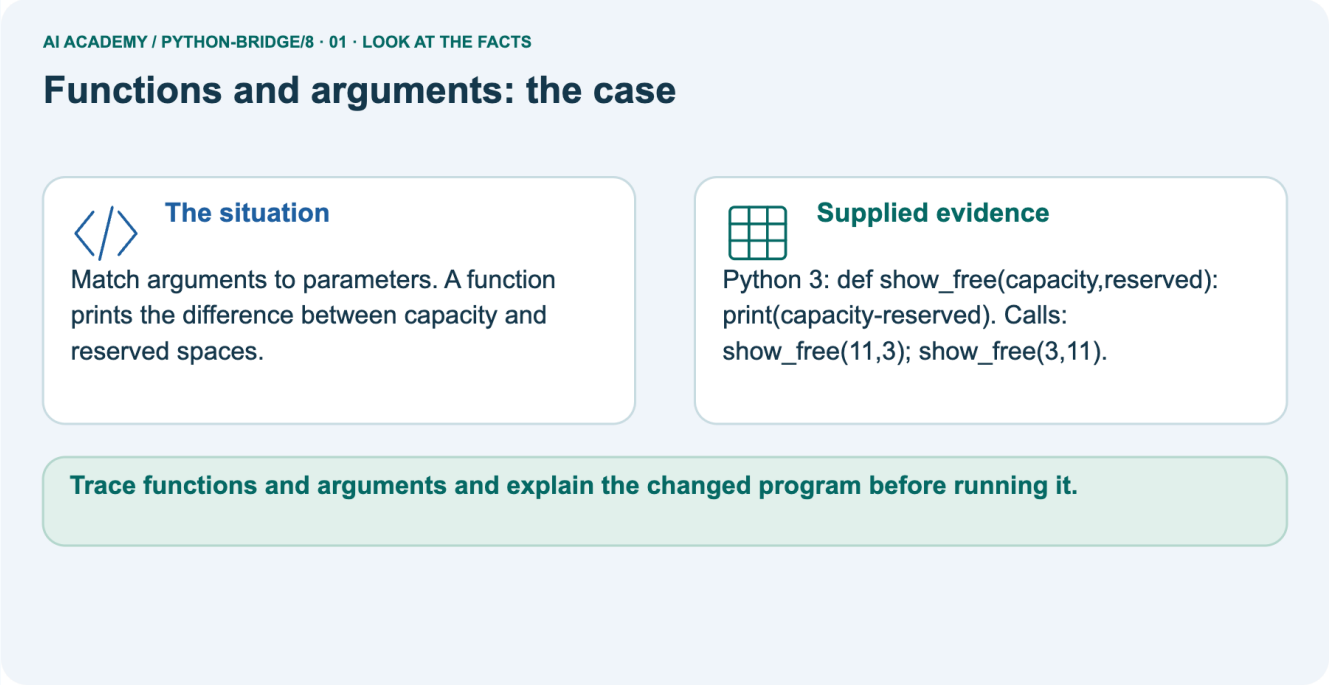
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

## Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

## 01. Functions and arguments: the case




Follow the labelled picture with the matching explanation. Open the vector diagram

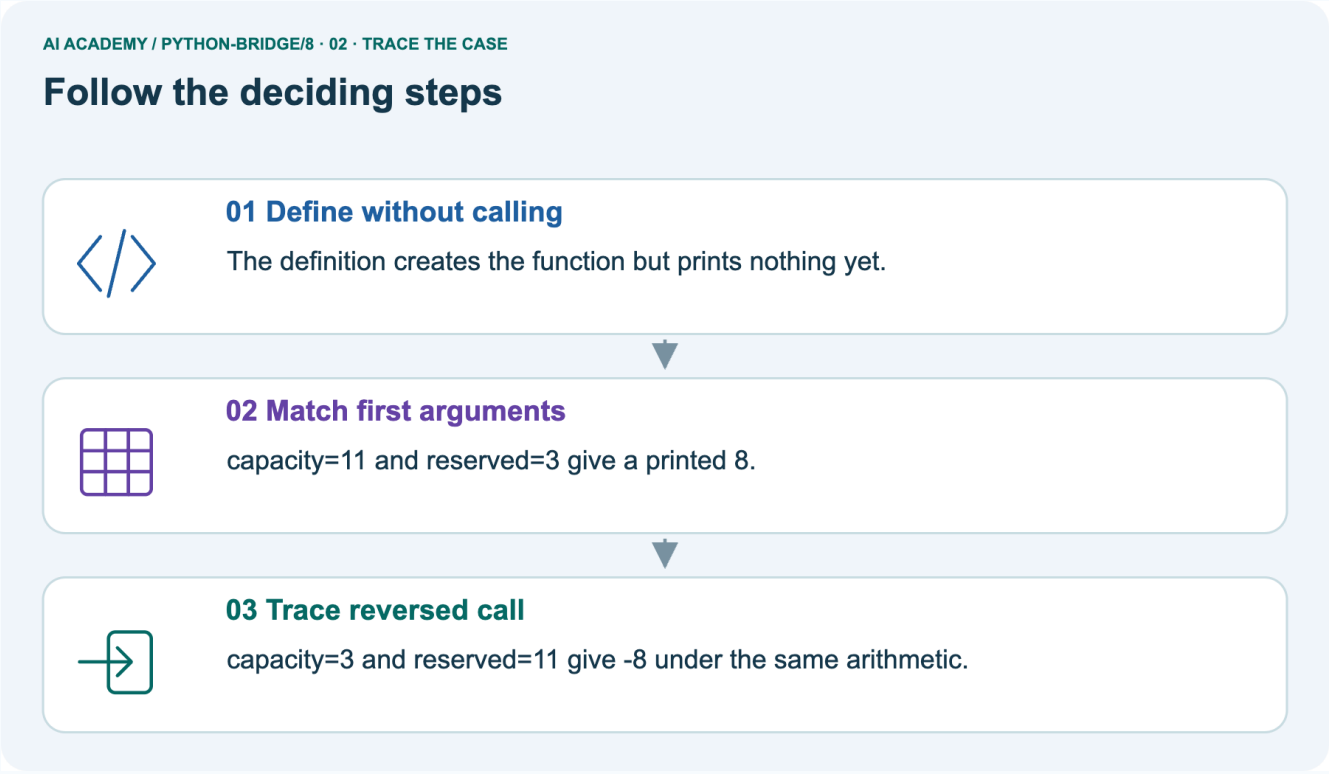
Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces. Supplied facts: Python 3: def show_free(capacity,reserved): print(capacity-reserved).

**Call**s: show_free(11,3); show_free(3,11).

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

Define without **call**ing: The **definition** creates the **function** but prints nothing yet. → Match first **argument**s: capacity=11 and reserved=3 give a printed 8. → Trace reversed **call**: capacity=3 and reserved=11 give -8 under the same arithmetic.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Arguments matched in order

AI ACADEMY / PYTHON-BRIDGE/8 · 03 · INSPECT THE LESSON MODEL

### Arguments matched in order

| Call | Parameter values | Printed output |
| --- | --- | --- |
| show_remaining(12, 4) | capacity 12; used 4 | 8 |
| show_remaining(10, 7) | capacity 10; used 7 | 3 |

Each call runs the same body with the values supplied for that call.

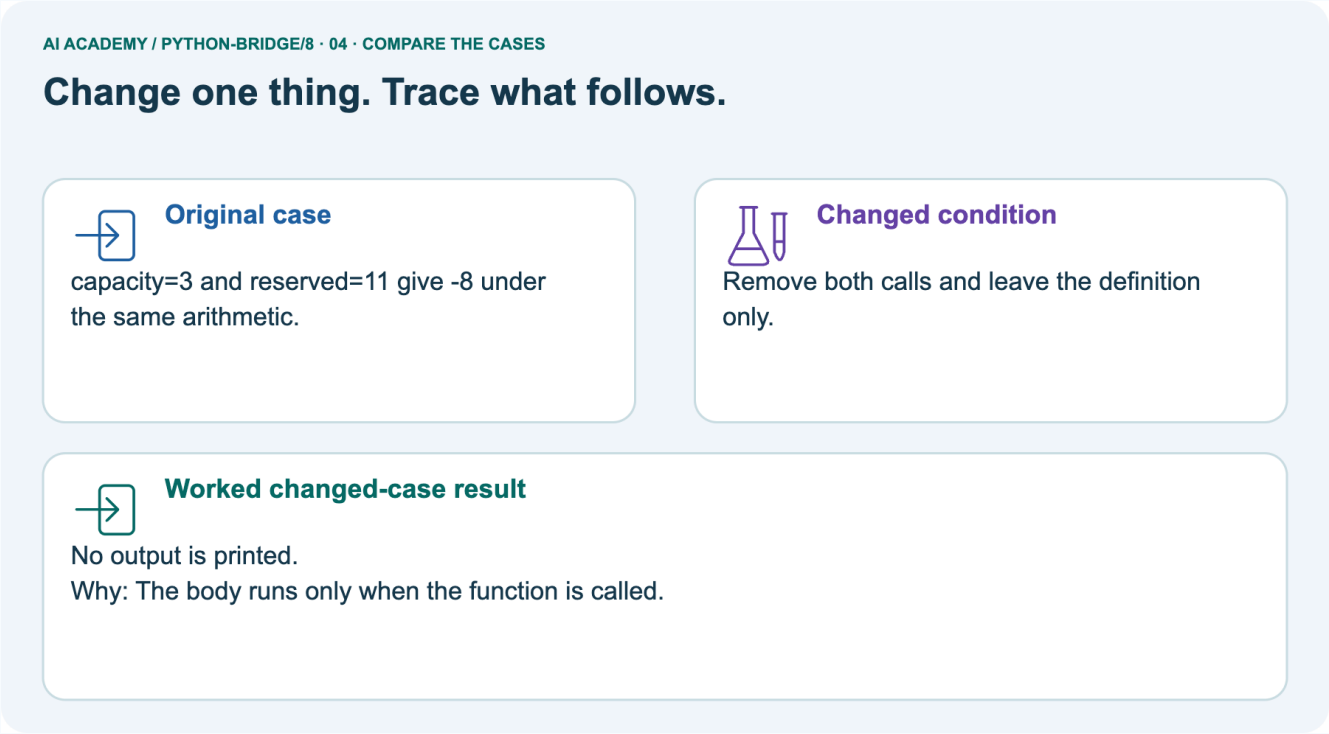
Follow the labelled picture with the matching explanation. Open the vector diagram

**Argument**s matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

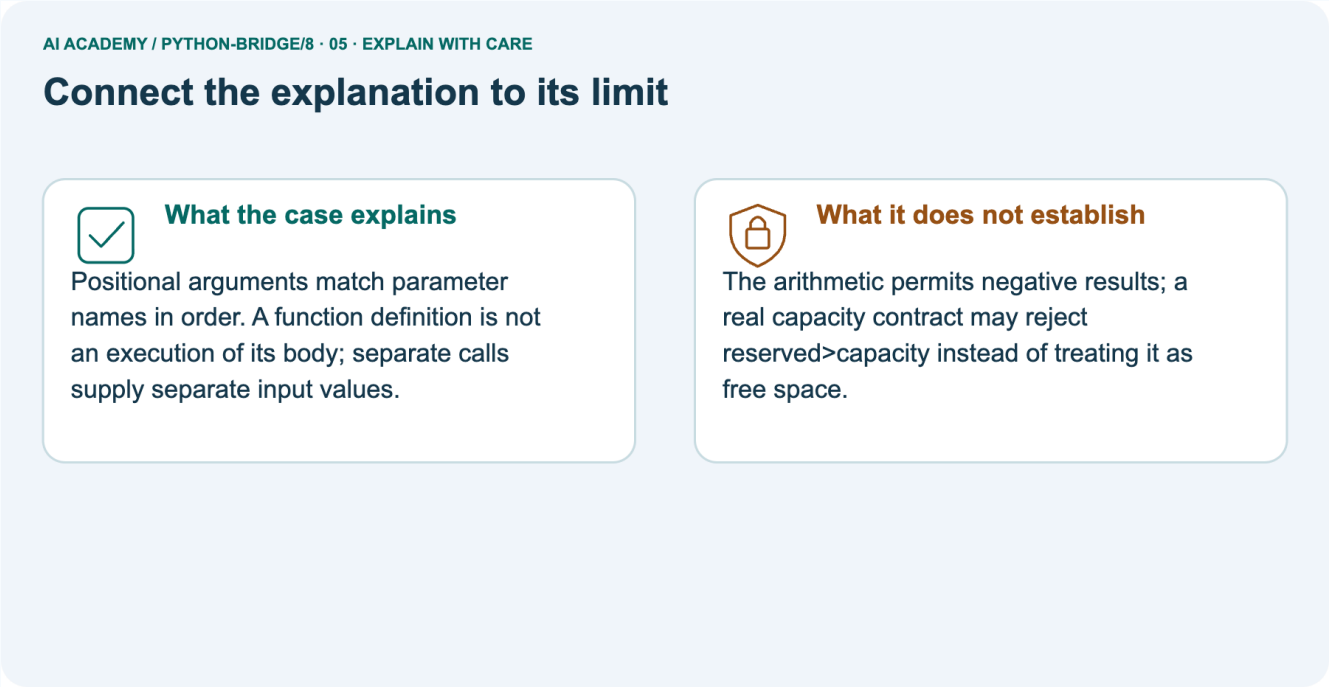
Original case: capacity=3 and reserved=11 give -8 under the same arithmetic. Changed condition: Remove both **call**s and leave the **definition** only. Expected result: No **output** is printed.

Reason: The body runs only when the **function** is **call**ed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

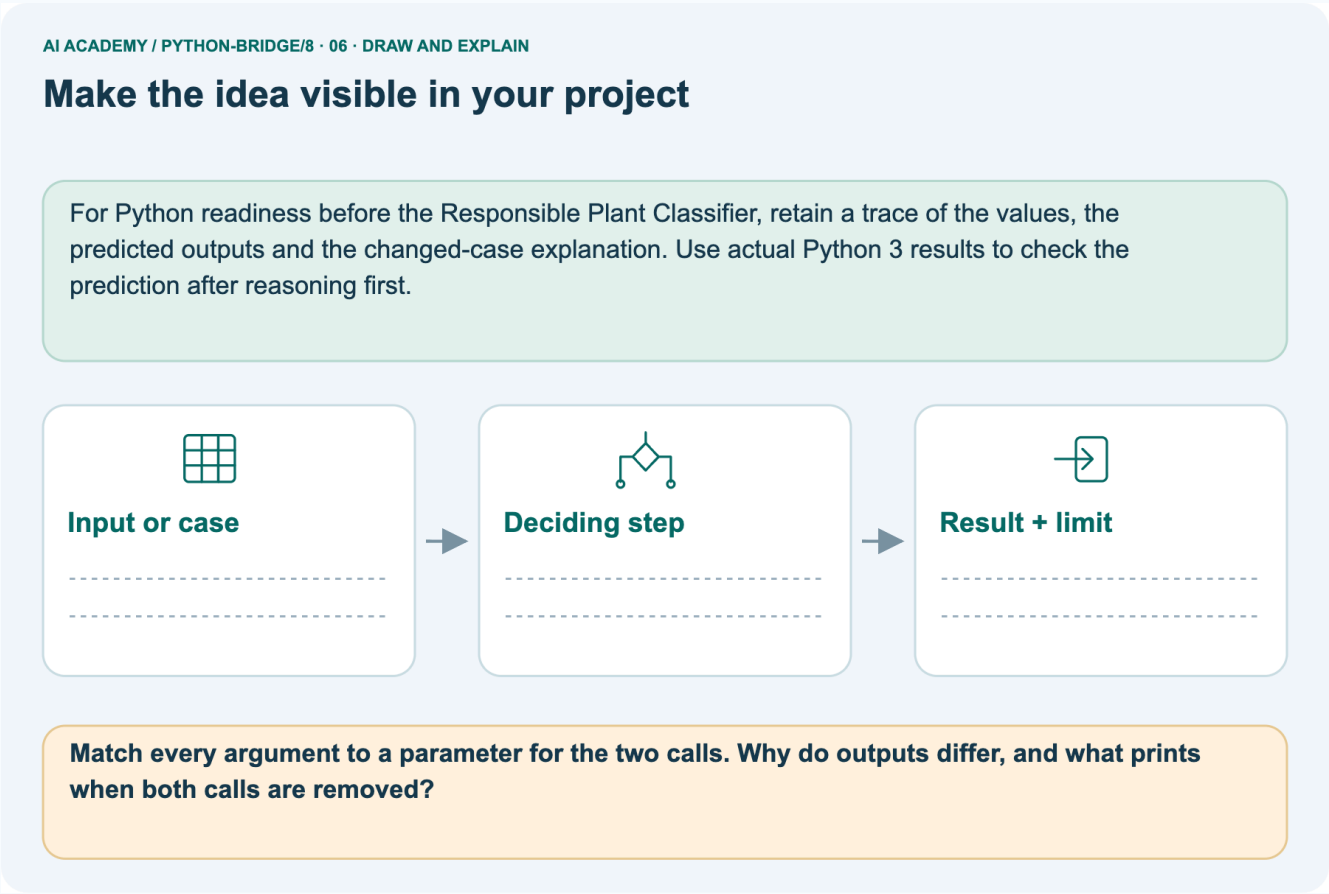
Positional **argument**s match **parameter** names in order. A **function definition** is not an execution of its body; separate **call**s supply separate **input** values.

Limit: The arithmetic permits negative results; a real capacity contract may reject reserved>capacity instead of treating it as free space.

Explain the case, then point to a limit on the claim.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Match every **argument** to a **parameter** for the two **call**s. Why do **output**s differ, and what prints when both **call**s are removed?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

**01. Functions and arguments: the case**

**02. Follow the deciding steps**

**03. Arguments matched in order**

**04. Change one thing. Trace what follows.**

**05. Connect the explanation to its limit**

**06. Make the idea visible in your project**

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.